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26 March 1984

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MILITARY STANDARD

FASTENER TEST METHODS, METRIC

METHOD 107

VIBRATION

FSC 53GP

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DEPARTMENT OF DEFENSE
Washington, DC 20301

Fastener Test Methods, Metric, Method 107, Vibration

DOD-STD-1312-107

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FOREWORD

This standard sets forth standard test procedures for determining the capability of externally threaded fasteners to withstand the accelerated vibration conditions on any fastener system capable of providing a clamp-up load.

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1. SCOPE

1.1 Applicability. This method establishes a standard procedure for conducting accelerated vibration tests on any fastener system capable of providing a clamp-up load. This method is intended to define the test required to determine the life and compare fasteners under vibratory conditions.

2. REFERENCED DOCUMENTS

Not applicable.

3. DEFINITIONS

Not applicable.

4. GENERAL REQUIREMENTS

4.1 Test apparatus.

4.1.1 Testing machine. The testing machine shall be capable of vibrating the fixture as shown on figure 1 with an essentially sinusoidal wave form at a frequency of 29 to 30 Hz with an amplitude of 11.05 to 11.81 mm. The machine shall be provided with a means to verify the proper amplitude, such as an optical wedge gage.

4.1.2 Test fixtures. Unless otherwise specified, the test fixture shall conform to figure 1. The fixtures are intended for the testing of threaded fasteners and are satisfactory for testing supplementary locking means such as cotter pins.

4.1.2.1 Other fixtures. Other fixtures may be used, provided the following criteria are met (see figure 1 for B, C and D):

a. The cross-section area of the cylinder shall be at least 1.5 times the area of the cylinder ID.

$$\frac{\pi}{4}\ (D^2 - C^2) \geq \frac{1.5\ \pi\ C^2}{4}$$

$$D \geq 1.58C$$

b. The slot clearance shall be equal to 18.9 mm.

$$B-D = 18.9 \text{ mm}$$

c. Sufficient clearance shall be provided between the cylinder and fixture so that the cylinder moves freely in the slot.

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d. The mass of cylinder assembly with test parts in place shall be distributed so that the cylinder axis stays perpendicular to the applied vibratory forces during the test. This is accomplished by adding washers.

4.2 Test specimen (sample size). The number of specimens to be tested shall be as specified in the acquisition document or product specification. The test specimen shall not be altered in any manner other than that described in the product design specification.

5. DETAIL REQUIREMENTS

5.1 Test procedures. Unless otherwise specified, the following procedure shall be followed:

a. Assemble all components on the test cylinder, using washers as required. Threaded parts shall be preloaded as required, using the appropriate seating torque as specified in the procurement document.

b. When specified, for baked parts use a separate spacer of comparable bolt material and length, seat nut against spacer and bake.

c. Disassemble and assemble with test cylinder to provide the reuse as specified. The last assembly shall be with the cylinder in the fixture.

d. If rotation is a measure of failure, scribe or otherwise mark parts to determine relative movement.

e. Lightly lubricate the sliding surfaces of the test fixture under the cylinder flanges and washers with SAE 20 oil so that the cylinder assembly will freely traverse the slots of the fixture.

f. Vibrate the fixture for the required time and inspect the components for failure.

g. Failure shall be indicated by the product design specification, normally as a minimum amount of rotation or the absence of cracking within a required time.

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6. NOTES

6.1 Test reports. The test report shall contain the following:

a. Description and part number of all components tested.

b. Installation data such as seating torque, bake time and temperature, and reuses.

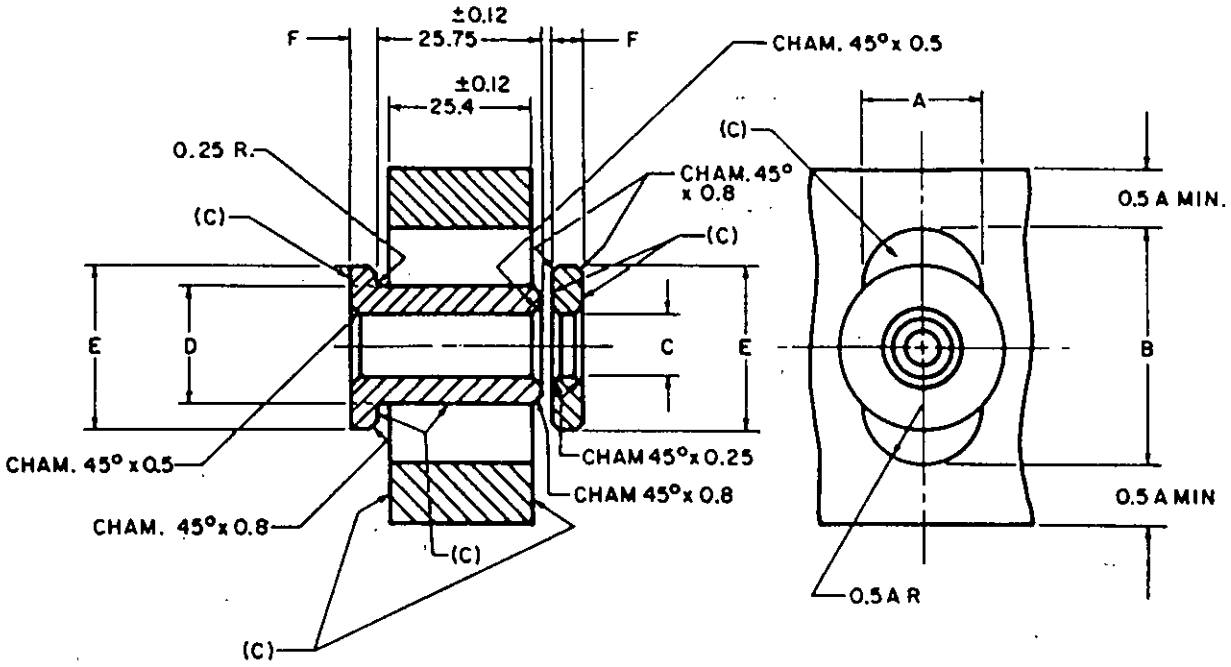
c. Test results.

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Air Force - 11
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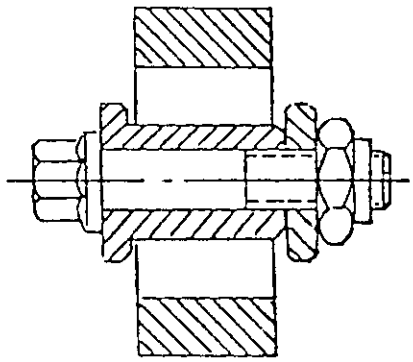
NOMINAL DIAMETER	SLOT WIDTH A +0.13 -0.05	SLOT HEIGHT B ±0.05	WASHER AND CYLINDER I.D. C ±0.05	CYLINDER O.D. D ±0.05	WASHER O.D. E ±0.10	WASHER FACE THICKNESS F ±0.05
5	8.6	27.3	5.2	8.4	14.1	3.0
6	11.9	30.7	6.3	11.7	18.0	4.1
7	13.8	32.5	7.3	13.6	19.5	4.1
8	15.9	34.6	8.3	15.7	22.3	4.1
10	19.6	38.3	10.4	19.4	26.0	4.1
12	24.2	43.4	12.4	24.5	33.0	4.7
14	28.2	46.9	14.4	28.0	39.0	5.4
16	31.8	50.5	16.4	31.6	43.3	5.4

a - Material: Steel.

b - Heat Treatment: Rc 40-45

c - Surface Texture: On surface coded (c) the finish is to be 0.8 micrometres max.

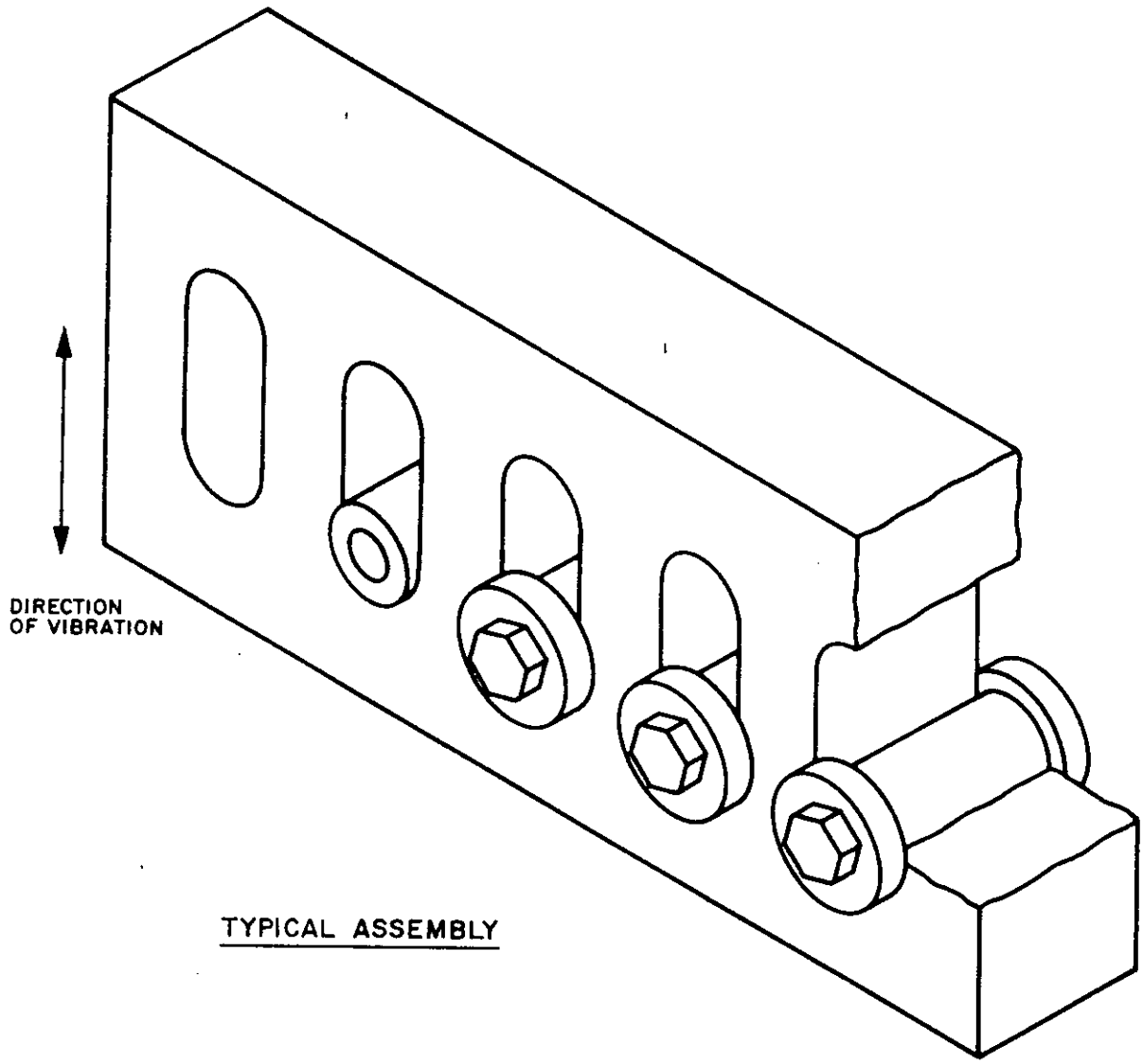
d - Use: This fixture is used primarily for vibration tests of self-locking bolts and self-locking nuts. An assembly of the fixtures, bolt and nut is shown.



e - All dimensions in millimeters.

FIGURE 1. Vibration fixture.

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TYPICAL ASSEMBLY

FIGURE 1. Vibration fixture (continued).

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a. Paragraph Number and Wording:

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